SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Application-Based Questions 3Questions)

Course Code:	ITA0609	Course Title:	Machine Learning
CO Assessed:	CO1 – Analyze (BL 3)	Assessment:	Assessment Tool 1 – Application Based Questions.
Weightage:	40% of CIA	Total Marks:	30 (3 Questions ×10 Mark)
CO1 Statement:	*To acquire a foundational understanding of key machine learning concepts including version spaces, candidate elimination, inductive bias, and heuristic search (BL1)*		
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Section / Batch:	SLOT A	Date:	13.7.26

Code of Conduct

I (N.kishan kumar-192424404) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

1. ML System Design — Predicting Heart Disease / Diabetes

1. Problem Definition

The goal is to build a binary/multi-class classification system that predicts the likelihood of a patient having a disease such as heart disease or diabetes, using patient records containing attributes like age, blood pressure, glucose level, BMI, cholesterol, and lifestyle habits (smoking, exercise, diet, alcohol use). Early prediction allows preventive treatment and reduces risk.

2. Concept Map — Overall Pipeline

Patient Data ➡ **Pre- processing** ➡ **Feature Selection** ➡ **Model Training** ➡ **Evaluation & Deployment**

3. Data Preprocessing Steps

- ✎ Handle missing values: impute numeric fields (age, BP, glucose) with mean/median; categorical lifestyle fields with mode.
- ✎ Remove/cap outliers (e.g. implausible BP or glucose readings) using IQR or z-score capping.
- ✎ Encode categorical variables (smoker: yes/no, diet type) using label or one-hot encoding.
- ✎ Scale numeric features (Min-Max or Z-score standardisation) since age, BP and glucose have very different ranges.
- ✎ Balance the classes with SMOTE/undersampling, since disease-positive cases are usually the minority class.

4. Feature Selection

- ✎ Correlation heat-map to drop highly redundant numeric features.
- ✎ Chi-square test for categorical lifestyle attributes vs the target label.
- ✎ Recursive Feature Elimination (RFE) or feature-importance from a tree-based model to rank predictors such as glucose level and BP, which are typically the strongest signals.

5. Algorithm Choice & Justification

Algorithm	Why suitable	Limitation
Logistic Regression	Simple, fast, gives interpretable odds — clinicians can see how each factor changes risk.	Struggles with non-linear relationships.
Random Forest / XGBoost	Handles non-linear interactions (e.g. age × glucose), robust to outliers, gives feature importance.	Less directly interpretable than logistic regression.
Support Vector Machine	Works well on smaller, high-dimensional clinical datasets.	Slower on very large datasets; needs careful kernel tuning.
Neural Network	Can capture complex patterns if enough data is available.	Needs large data, acts as a 'black box', harder to justify to doctors.

★ Recommended approach:
Use Random Forest / Gradient Boosting as the main model for accuracy and built-in feature importance, alongside Logistic Regression as an interpretable baseline that clinicians can trust. This combination balances accuracy with the explainability that medical predictions require.

6. Model Training Steps

✎ Split data 80:20 (train:test) using stratified sampling so both classes are represented proportionally.

✎ Apply k-fold (e.g. 5-fold) cross-validation on the training set to get a stable performance estimate.

✎ Tune hyperparameters (tree depth, number of estimators, regularisation strength) with GridSearchCV / RandomizedSearchCV.

✎ Apply regularisation (L1/L2) or pruning to prevent overfitting on the relatively small medical dataset.

7. Evaluation for Reliable Predictions

✎ Report Accuracy, Precision, Recall (Sensitivity) and F1-score — Recall matters most, since missing a true disease case (false negative) is far costlier than a false alarm.

✎ Plot the ROC curve and compute AUC to judge overall discriminative power.

✎ Use a confusion matrix to see exactly which class is being mis-predicted.

✎ Before deployment, use SHAP/feature-importance plots so predictions stay explainable to medical staff, and monitor/retrain the model periodically as new patient data arrives.

Find-S Algorithm — Loan Approval Dataset

Given Dataset

Example	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

Find-S Algorithm — Concept

✱ Find-S only looks at POSITIVE examples (Loan Approved = Yes) and ignores every negative example.

✱ It starts from the most specific hypothesis h_0 = (∅, ∅, ∅, ∅) and, for each positive example, generalises only the attributes that don't match — replacing them with '?'.

Concept Map — Hypothesis Generalisation

h_0 (∅,∅,∅,∅) ➡ **h_1 ← Ex.1 (Emp,High, Good,Clean)** ➡ **h_2 ← Ex.2 (Emp,?, Good,Clean)** ➡ **h_4 ← Ex.4 (Emp,?,?, Clean)**

Step-by-Step Trace

Step	Example used	Label	Action	Hypothesis after step
0	—	—	Initialise	(∅, ∅, ∅, ∅)
1	Ex.1: Employed,High,Good,Clean	Yes	Take values as-is	(Employed, High, Good, Clean)
2	Ex.2: Employed,Medium,Good,Clean	Yes	Salary differs (High vs Medium) → generalise to ?	(Employed, ?, Good, Clean)
3	Ex.3: Unemployed,Low,Poor,Bad	No	Ignored (negative example)	(Employed, ?, Good, Clean)
4	Ex.4: Employed,Medium,Excellent,Clean	Yes	Credit Score differs (Good vs Excellent) → generalise to ?	(Employed, ?, ?, Clean)

✓ Final Hypothesis: h = (Employed, ?, ?, Clean)

Interpretation: based on the given examples, a loan is approved whenever the applicant is Employed and has a Clean loan history — Salary level and Credit Score do not matter, since both attributes were generalised away.